

More about Vectors

Vectors: Abstract View:



- Collection of elements: (Think of them as a grocery bag)

$$\begin{bmatrix} 3 \text{ apples} \\ 2 \text{ oranges} \end{bmatrix}; \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

- Things you CAN do with vectors:
 - Add them: $\begin{bmatrix} 3 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$ (combine two grocery bags)
 - Multiply by a scalar: $2 \times \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$ (double your order)

More about Vectors

Remember, whenever we learn about a new object in this course, we want to understand it from three modes of thinking:

- Computational
- Geometric
- Algebraic

Vectors: Algebraic view

$$\mathbb{R}^2 \quad \mathbb{R}^3$$

Column Vector: A list of real numbers with only one column is called a column vector, or simply a vector.

We will use the notation $\mathbb{R}^d$ to represent the set of all the vectors that have exactly ***d*** components.

Euclidean Space: A set of all vectors with ***d*** entries is known as a Euclidean Space of dimension ***d*** and is denoted by $\mathbb{R}^d$.

Examples: Vectors in $\mathbb{R}^2$:

$$a = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad b = \begin{bmatrix} -3 \\ -2 \end{bmatrix} \quad c = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Vector Notation:

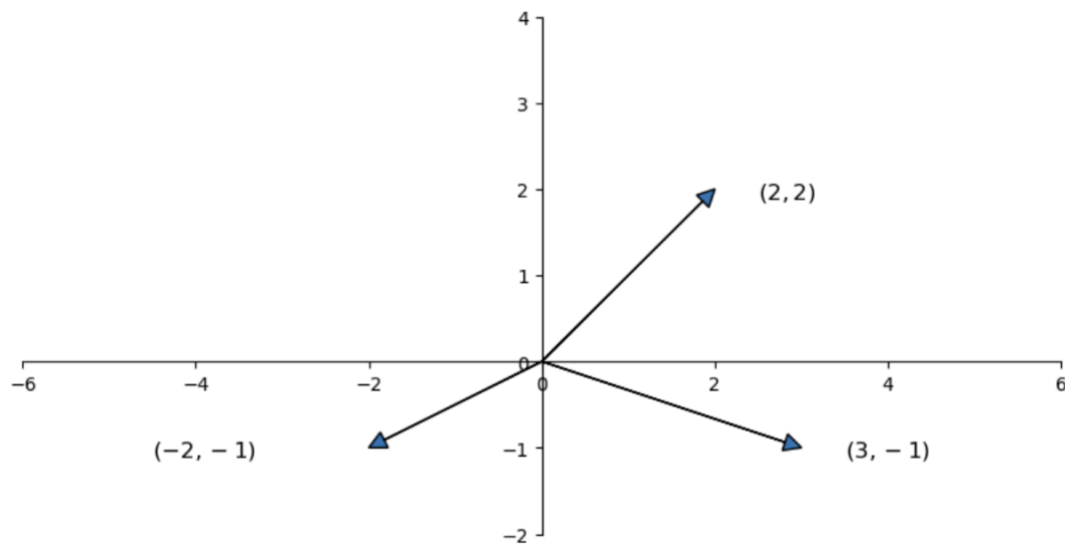
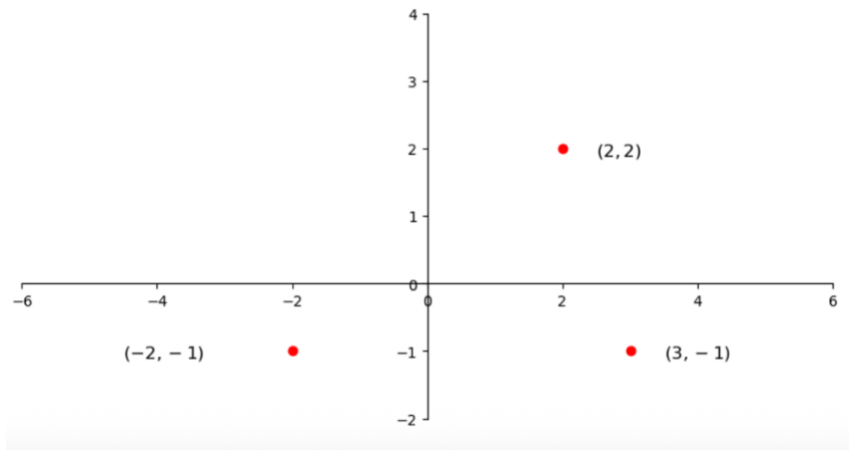
Notation	Example	Notes
Column	$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$	Standard in Linear Algebra
Row	$[2 \quad 3 \quad 4]$	Transpose of column vectors
Angle Brackets	$\langle 2, 3, 4 \rangle$	Common in physics
Tuple	$(2, 3, 4)$	Often used to represent points

Geometric Representation

Geometrically, a vector can be thought of as a point in $\mathbb{R}^d$.

Both $\begin{bmatrix} -2 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} -2 & -1 \end{bmatrix}$ can be viewed as a point on a plane.

Equivalently, we can represent a vector by drawing an arrow from the origin to the point.



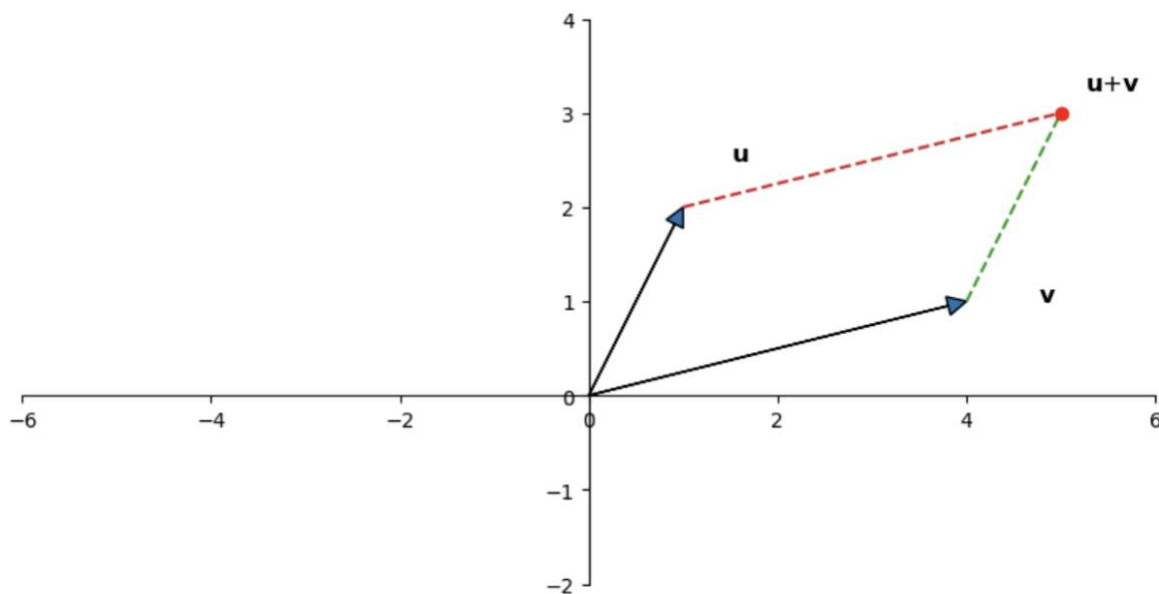
Vector Manipulations:

- *Vector Equality*: Two vectors of the same size are equal if and only if their corresponding entries are equal.
- *Scalar Multiplication*: Multiplying a vector by a scalar is accomplished by multiplying each entry by the scalar.
- *Vector Addition*: If two vectors are the same size, then their sum is another vector whose entries are the corresponding sums.

Vector Addition: Geometric View

A geometric interpretation of vector sum is as a **parallelogram**.

If $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^2$ are represented as points in the plane, then $\mathbf{u} + \mathbf{v}$ corresponds to the fourth vertex of the parallelogram whose other vertices are $\mathbf{u}$, $\mathbf{0}$ and $\mathbf{v}$.

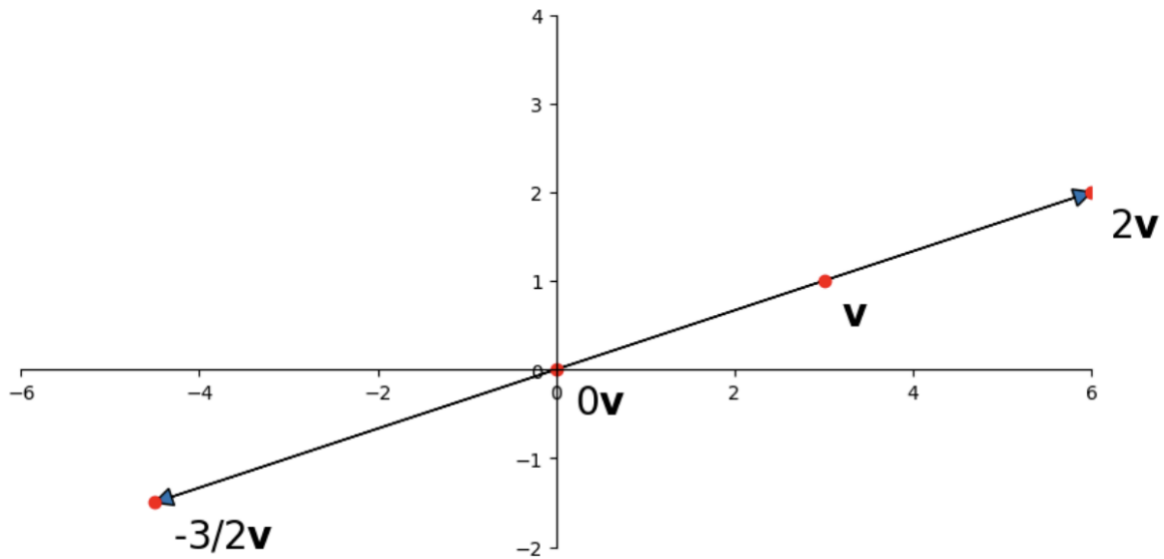


Scalar Multiplication: Geometric View

For a given vector $\mathbf{v}$ and a scalar $a \in \mathbb{R}$, multiplying a and $\mathbf{v}$ corresponds to lengthening $\mathbf{v}$ by a factor of a .

So, $2\mathbf{v}$ is twice as long as $\mathbf{v}$.

Multiplying by a negative value reverses the “direction” of $\mathbf{v}$.



Linear Combinations

Applying vector addition and scalar multiplication to vectors produces their linear combination.

Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_p$ be vectors in $\mathbb{R}^n$. Let $c_1, c_2, c_3, \dots, c_p$ be scalars. Then, a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_p$ is given by $\mathbf{y} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \dots + c_p\mathbf{v}_p$.

The c_i values are called weights. Weights can be any real number, including zero.

Examples:

$$3\mathbf{v}_1 + \mathbf{v}_2, \quad \begin{pmatrix} 1 \\ 2 \end{pmatrix} \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \mathbf{v}_1 + 0\mathbf{v}_2, \quad 0 = 0\mathbf{v}_1 + 0\mathbf{v}_2$$

Linear Combinations: Geometric Representation

Suppose $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

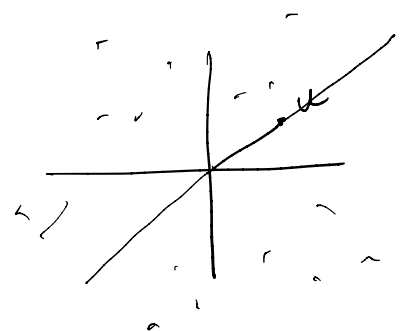
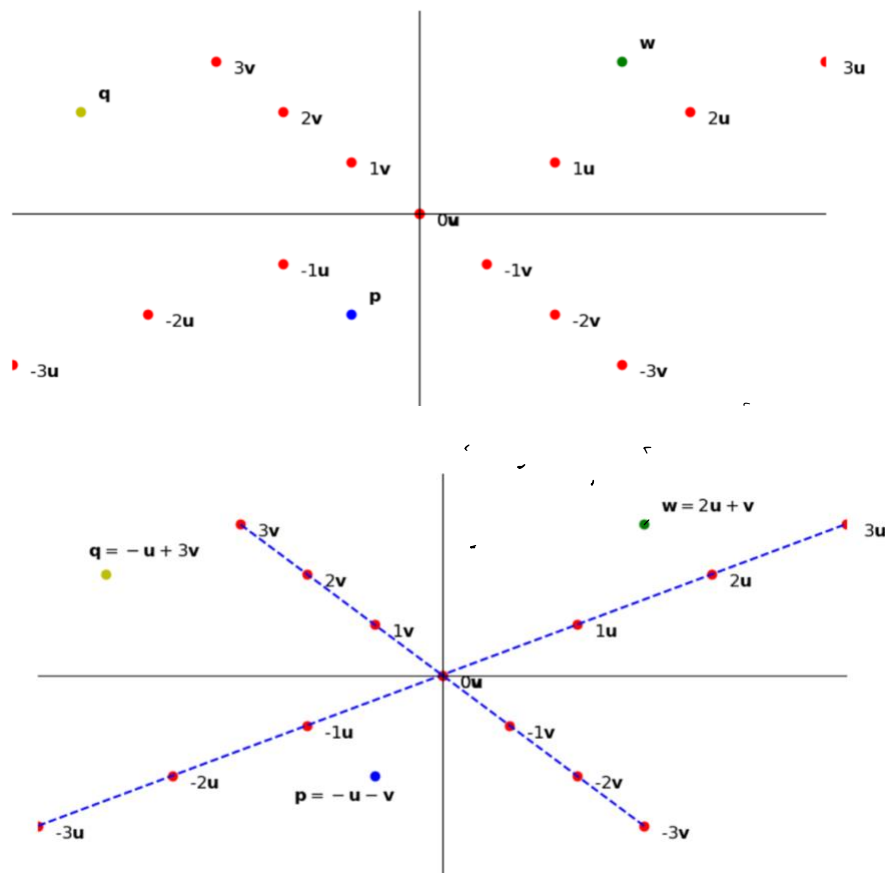
Then the linear combination $c\mathbf{u} + d\mathbf{v}$ corresponds to a walk in $\mathbf{u}$ direction followed by a walk in $\mathbf{v}$ direction.

Let's determine the locations of the following vectors geometrically:

$$\mathbf{w} = 2\mathbf{u} + \mathbf{v}$$

$$\mathbf{p} = -\mathbf{u} - \mathbf{v}$$

$$\mathbf{q} = -\mathbf{u} + 3\mathbf{v}$$

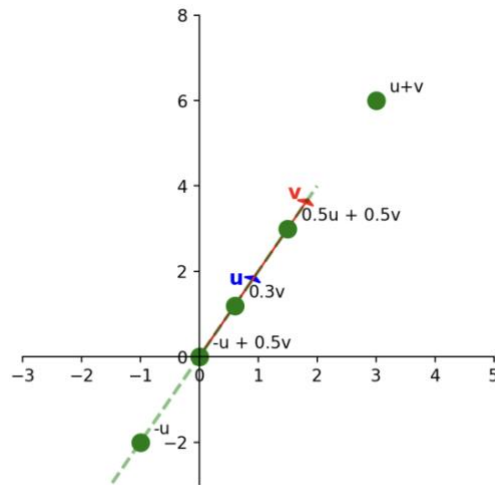


Can we reach any point in $\mathbb{R}^2$ as a linear combination of two vectors?

Collinear vs Non-Collinear vectors

Collinear vectors: Think: Stuck on the same line

Example: $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ Note: $\mathbf{v} = 2\mathbf{u}$



Non-Collinear vectors:

Example: $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ Note: These point in different directions

How would linear combinations of $\mathbf{u}$ and $\mathbf{v}$ look? Think: $\mathbf{u} + \mathbf{v}$ or $\mathbf{u} + 2\mathbf{v}$

Why does this matter?

We need vectors that are **linearly independent** (not scalar multiples of each other) to span the full space.

More about this later!!

Linear combinations in 3-D

Let $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be three vectors in $\mathbb{R}^3$ and none of them is a linear combination of the other(s).

- What is the picture of all linear combinations of $\mathbf{u}$?
- What is the picture of all linear combinations of $\mathbf{u}$ and $\mathbf{v}$?
- What is the picture of all linear combinations of $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$?

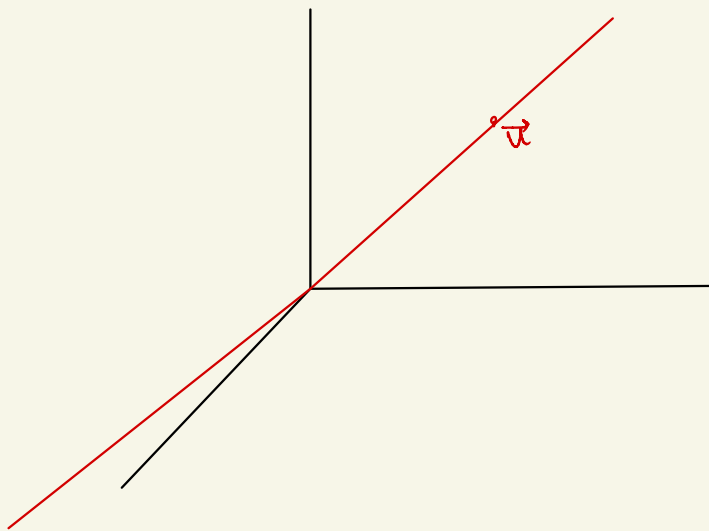
* What happens when $\vec{u}$ is a zero vector?

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

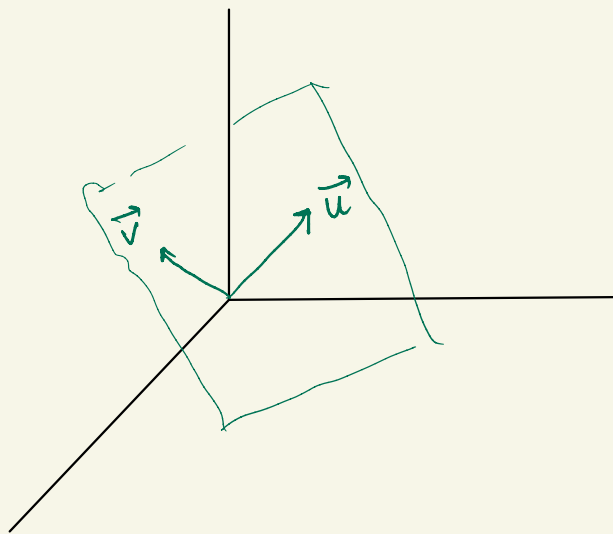
1. All linear combinations of $\vec{u}$.

→ A line through the origin



2. All linear combinations of $\vec{u}$ and $\vec{v}$

→ A plane through the origin



3. All linear combinations of $\vec{u}, \vec{v}, \vec{w}$

→ all of $\mathbb{R}^3$